

cGMP-AMP synthase (cGAS) is a cytosolic DNA sensor that becomes activated upon binding DNA. Activated cGAS produces cGAMP, a second messenger that activates a protein known as STING, a downstream protein. STING activation causes recruitment of interferon regulatory factor-3 (IRF3), leading to inflammation and cytokine release. Researchers investigated a potential role in cell immortalization (extended proliferation) for cGAS in mouse embryonic fibroblasts (MEFs) and human tumor cells.

Experiment 1

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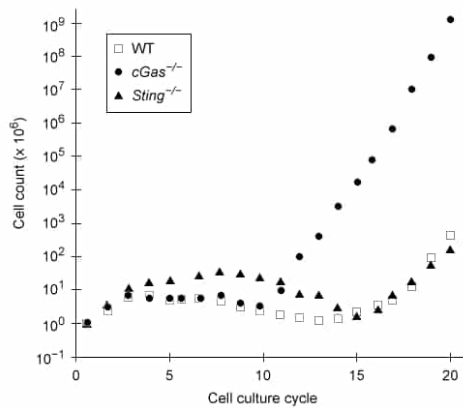


Figure 1 Effect of deletion of cGAS or STING on immortalization in MEFs

Experiment 2

The role of cGAS in MEF cell senescence was tested by staining $cGas^{-/-}$ and WT cells with SA- β -Gal, a commonly used method to detect senescent cells. The percentage of SA- β -Gal positive cells during 20 cycles of cell culture was measured in both cell types (Figure 2).

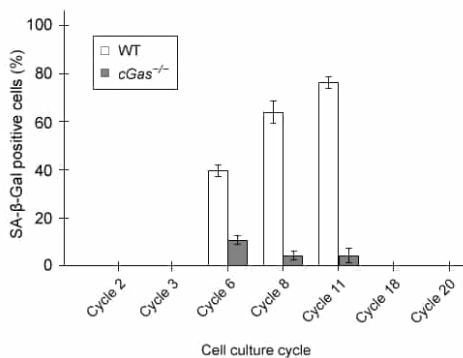


Figure 2 Percentage of $cGas^{-/-}$ and WT cells positive for SA- β -Gal

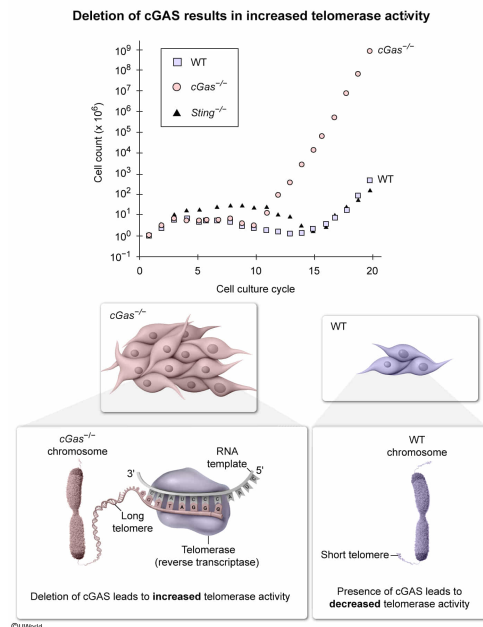
Compared to WT cells, cells with deletion of cGAS are likely to exhibit:

- ☐ A. increased apoptosis. [5%]
- ☐ B. decreased cell immortalization. [12%]
- ☒ C. increased telomerase activity. [67%]
- ☐ D. decreased telomere length. [15%]

Correct

67% answered correctly

Explanation:



All organisms undergo **senescence**, a natural aging process during which cells and tissues acquire damage (eg, **mutations**). Over time, this aging process can cause cells to diminish in function.

In cells, senescence occurs when progression of the **cell cycle** ceases to occur. Senescence may result from the degradation of chromosome ends (ie, **telomeres**) that occurs over time because **DNA replication** processes **do not allow** complete replication of linear chromosome ends. However, some cells (eg, stem cells) express an enzyme known as **telomerase**, a reverse transcriptase that allows synthesis of chromosome ends and the prevention of senescence (ie, cell immortalization).

In some cases, senescent cells are eliminated (ie, via **programmed cell death**), but the effects of senescence can accumulate over time. This eventually leads to decreased organism function (eg, reproduction).

In this passage, cGAS is a DNA sensor being investigated for a potential role in cell immortalization in mouse embryonic fibroblasts (MEFs). Figure 1 shows the cell count after each of 20 cycles of cell culture in wild-type (WT) MEFs and in MEFs lacking cGAS (cGas^{-/-}). While the cell count for WT cells and cGas^{-/-} cells is quite similar for cell culture cycles 1-10, cGas^{-/-} cells demonstrate much higher cell counts than WT cells for cycles 11-20.

Therefore, deletion of cGAS produces cells that reproduce at higher rates; in other words, this provides support for the idea that deletion of cGAS results in increased (*not* decreased) cell immortalization compared to WT cells (**Choice B**). Cells that undergo immortalization likely demonstrate increased (*not* decreased) telomere length (**Choice D**) due to **increased telomerase activity**.

(Choice A) Increased **apoptosis** is a hallmark of senescence, not cell immortalization. Therefore, because deletion of cGAS results in increased cell immortalization, these cells show a decreased (*not* increased) apoptosis.

Educational objective:

Senescence is a natural aging process during which cells acquire damage and eventually diminish in function over time. Some senescent cells may be eliminated from organisms; however, over time, the accumulating effects of senescence can cause a decline in organism functions.

Time spent: 0 sec
Subject:
Biology

QID: 403804
System:
Reproduction

cGMP-AMP synthase (cGAS) is a cytosolic DNA sensor that becomes activated upon binding DNA. Activated cGAS produces cGAMP, a second messenger that activates a protein known as STING, a downstream protein. STING activation causes recruitment of interferon regulatory factor-3 (IRF3), leading to inflammation and cytokine release. Researchers investigated a potential role in cell immortalization (extended proliferation) for cGAS in mouse embryonic fibroblasts (MEFs) and human tumor cells.

Experiment 1

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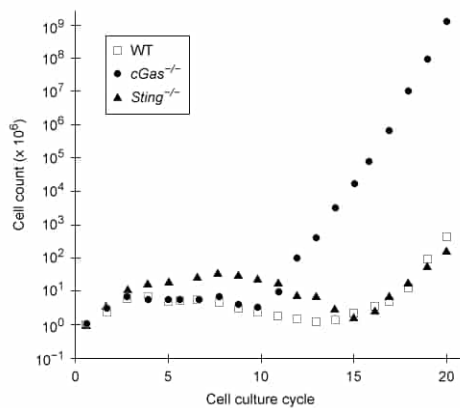


Figure 1 Effect of deletion of cGAS or STING on immortalization in MEFs

Experiment 2

The role of cGAS in MEF cell senescence was tested by staining $cGas^{-/-}$ and WT cells with SA-β-Gal, a commonly used method to detect senescent cells. The percentage of SA-β-Gal positive cells during 20 cycles of cell culture was measured in both cell types (Figure 2).

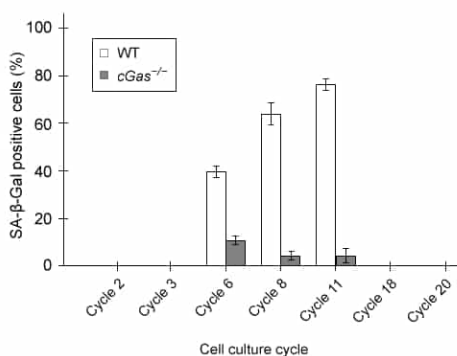


Figure 2 Percentage of $cGas^{-/-}$ and WT cells positive for SA-β-Gal

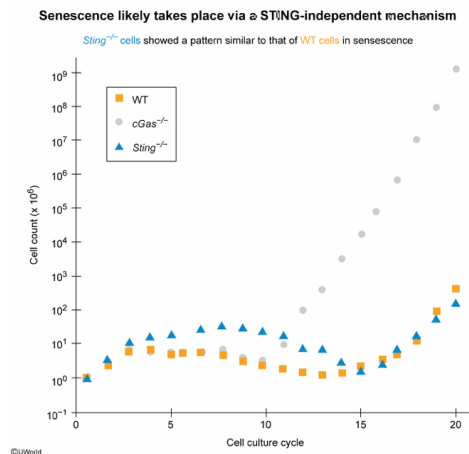
Do the results in Figure 1 support the hypothesis that STING is involved in cell immortalization?

- ☐ A. Yes; WT cells demonstrated significantly decreased immortalization compared to $Sting^{-/-}$ cells. [3%]
- ☐ B. Yes; cell senescence is directly proportional to the amount of STING expressed in cells. [4%]
- ☐ C. No; the presence of STING did not affect immortalization of cells, suggesting that cells expressing STING demonstrate decreased inflammation. [22%]
- ☒ D. No; STING deletion had no effect on cell senescence, suggesting the senescence pathway most likely takes place via a STING-independent mechanism. [69%]

Correct

60% answered correctly

Explanation:



Cell immortalization occurs when cells adapt to aging in a way that allows them to continue dividing indefinitely, unlike typical (mortal) cells. The passage states that cGAS, a DNA sensor, is activated upon binding DNA. Once activated, cGAS produces the **second messenger** cGAMP, which serves to activate the STING protein. STING activation leads to inflammation and **cytokine** release.

This question asks whether the results in Figure 1 support the hypothesis that STING is involved in cell immortalization. Figure 1 shows the results of an experiment in which wild-type (WT) mouse embryonic fibroblast cells (MEFs) and MEFs lacking STING underwent 20 cycles of cell culture. The cell count was measured at the end of each cycle. Throughout the experiment, the cell count in cells lacking STING and in WT cells was generally equivalent. This suggests that **STING deletion** had **no effect** on **cell senescence**. Therefore, the senescence pathway in cells most likely takes place via a **STING-independent mechanism**.

(Choices A and B) The results in Figure 1 do *not* support the hypothesis that STING is involved in cell immortalization. Furthermore, WT cells did *not* demonstrate significantly decreased immortalization compared to *Sting*^{-/-} cells, and cell senescence is *not* directly proportional to the amount of STING expressed in cells.

(Choice C) While the presence of STING did not affect immortalization in cells, the passage states that STING activation leads to inflammation and cytokine release. Therefore, cells expressing STING likely demonstrate increased (*not* decreased) inflammation.

Educational objective:

Cell immortalization occurs when aging cells are able to continue dividing indefinitely.

Time spent: 0 sec

Subject:
Biology

QID: 403805

System:
Reproduction

cGMP-AMP synthase (cGAS) is a cytosolic DNA sensor that becomes activated upon binding DNA. Activated cGAS produces cGAMP, a second messenger that activates a protein known as STING, a downstream protein. STING activation causes recruitment of interferon regulatory factor-3 (IRF3), leading to inflammation and cytokine release. Researchers investigated a potential role in cell immortalization (extended proliferation) for cGAS in mouse embryonic fibroblasts (MEFs) and human tumor cells.

Experiment 1

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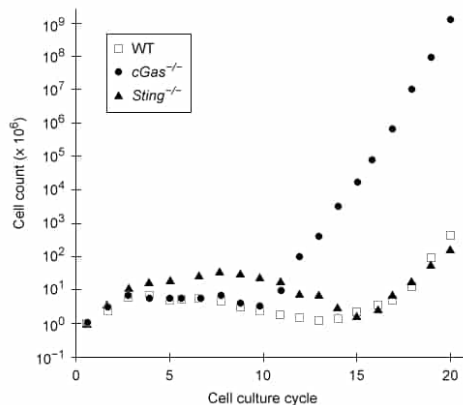


Figure 1 Effect of deletion of cGAS or STING on immortalization in MEFs

Experiment 2

The role of cGAS in MEF cell senescence was tested by staining *cGas*^{-/-} and WT cells with SA-β-Gal, a commonly used method to detect senescent cells. The percentage of SA-β-Gal positive cells during 20 cycles of cell culture was measured in both cell types (Figure 2).

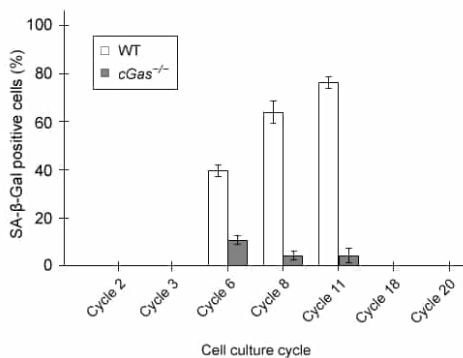


Figure 2 Percentage of *cGas*^{-/-} and WT cells positive for SA-β-Gal

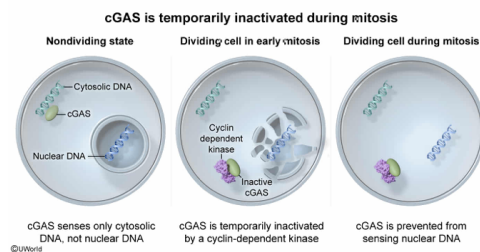
Researchers have noted that cGAS associates with a cyclin-dependent kinase during mitosis, causing cGAS to temporarily become inactive. The best explanation for this temporary inactivation would be:

- ✓ ☒ A. to prevent cGAS from sensing nuclear DNA. [51%]
- ☐ B. to prevent cell cycle checkpoints from taking place. [17%]
- ☐ C. to prevent cGAS from associating with nuclear DNA during the G₀ phase. [21%]
- ☐ D. to prevent nondisjunction during gamete formation. [9%]

Correct

52% answered correctly

Explanation:



The process by which cells replicate their content and divide is controlled by **checkpoints** within the **cell cycle**. These checkpoints ensure that the steps of the cell cycle, up until these points, have been executed in the correct manner. Checkpoints are regulated primarily by molecules called **cyclins** and cyclin-dependent kinases. If cells have abnormalities that cannot be repaired, these cells will undergo **apoptosis** rather than completing the cell cycle.

In this question, researchers have noted that cGAS associates with a cyclin-dependent kinase during mitosis, causing cGAS to temporarily become inactive. The passage states that cGAS is a cytosolic DNA sensor. This means that when the cell is performing its normal duties, cGAS would not sense nuclear DNA. However, during mitosis, the nuclear envelope disintegrates, exposing nuclear DNA that is not normally encountered by cGAS. Therefore, the association of cGAS with a cyclin-dependent kinase and the subsequent temporary inactivation of cGAS likely **prevents cGAS from sensing nuclear DNA** during mitosis.

(Choice B) Cell cycle checkpoints are critical for the health of an organism. Therefore, inactivation of cGAS to prevent cell cycle checkpoints from taking place would *not* be beneficial and could lead to cancer.

(Choice C) G_0 is a state of cell cycle arrest, in which a cell is not actively preparing to divide but is performing other essential functions. During the G_0 phase, the nuclear envelope is intact; therefore, cytosolic cGAS would *not* encounter nuclear DNA.

(Choice D) Gamete formation is the result of **meiosis**, *not* mitosis.

Educational objective:

The cell cycle is controlled by various checkpoints, which are regulated by cyclins and cyclin-dependent kinases, within the process to ensure that the steps of the cell cycle are executed in the proper manner.

Time spent: 0 sec

Subject:

Biology

QID: 403806

System:

Reproduction

cGMP-AMP synthase (cGAS) is a cytosolic DNA sensor that becomes activated upon binding DNA. Activated cGAS produces cGAMP, a second messenger that activates a protein known as STING, a downstream protein. STING activation causes recruitment of interferon regulatory factor-3 (IRF3), leading to inflammation and cytokine release. Researchers investigated a potential role in cell immortalization (extended proliferation) for cGAS in mouse embryonic fibroblasts (MEFs) and human tumor cells.

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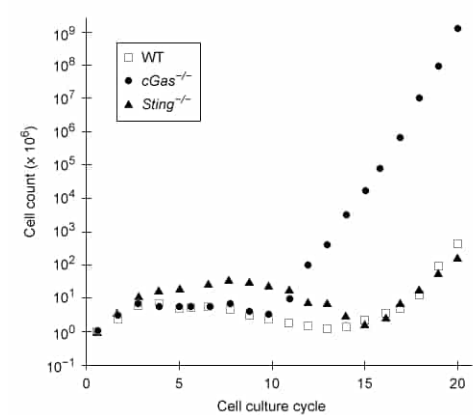


Figure 1 Effect of deletion of cGAS or STING on immortalization in MEFs

Experiment 2

The role of cGAS in MEF cell senescence was tested by staining *cGas*^{-/-} and WT cells with SA-β-Gal, a commonly used method to detect senescent cells. The percentage of SA-β-Gal positive cells during 20 cycles of cell culture was measured in both cell types (Figure 2).

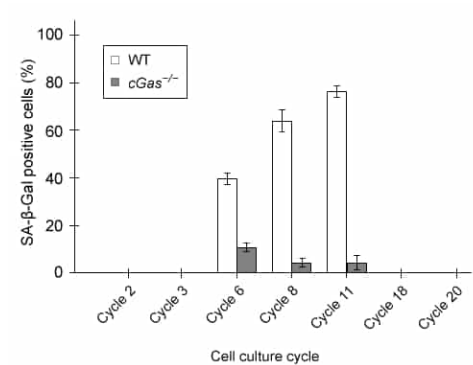
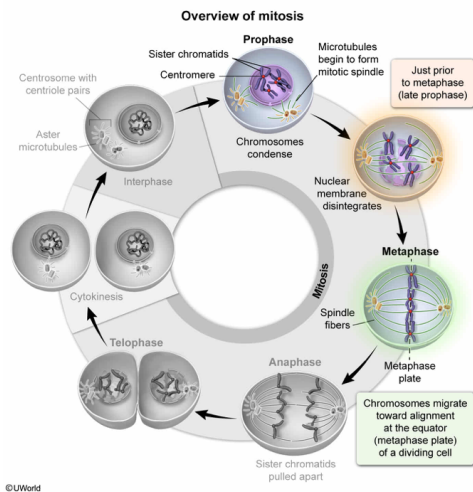


Figure 2 Percentage of *cGas*^{-/-} and WT cells positive for SA-β-Gal

To further elucidate the role of IRF3, researchers developed a protocol to synchronize replicating cells just prior to metaphase, allowing direct comparison of cellular IRF3 levels during mitosis. Which characteristic best describes the cells used in the experiment? Each cell displays:

- ☐ A. two centrosomes. [21%]
- ☐ B. a chromosome number that is double the diploid number for these cells. [20%]
- ☒ C. migration of chromosomes toward alignment at the equator of the dividing cells. [56%]
- ☐ D. the beginning of cleavage furrow formation. [2%]

Explanation:



Mitosis is the division of the nucleus and chromosomes during the **cell cycle**. In eukaryotic cells, mitosis consists of four phases:

1. **Prophase**: characterized by condensation of chromatin and nuclear envelope disintegration
2. **Metaphase**: in which chromosomes align on the metaphase plate (central region of the dividing cell) and become attached to spindle fibers
3. **Anaphase**: in which shortening of spindle fibers causes sister chromatids to separate and migrate away from one another
4. **Telophase**: characterized by reversal of condensation of chromatin and nuclear envelope reformation

The passage states that activated cGAS produces the **second messenger** cGAMP, which activates STING. This leads to inflammation and **cytokine** release via recruitment of IRF3. In this question, researchers developed a protocol to synchronize replicating cells just prior to metaphase to further elucidate the role of IRF3. This allows the researchers to directly compare cellular IRF3 levels during mitosis.

Because the experimental cells were synchronized just prior to metaphase, they would be characterized by **migration of chromosomes** toward **alignment at the equator** (ie, metaphase plate) of the dividing cells.

(Choice A) A cell undergoing mitosis displays two centrosomes at *each* phase. Consequently, the presence of two centrosomes would *not* in itself signify that a cell is just prior to metaphase.

(Choice B) After sister chromatids split in anaphase and prior to cytokinesis, the chromosome number is double the diploid number for replicating cells. However, in cells just prior to metaphase, the chromosomes have replicated (ie, during the **S phase** of interphase) but have not yet split; therefore, the chromosome number is still *equal* to the diploid number for these cells.

(Choice D) The cleavage furrow forms in cells at the beginning of **cytokinesis**, which typically overlaps partially with telophase. This occurs *after* the cells have gone through metaphase.

Educational objective:

Mitosis typically consists of four phases. During prophase, the nuclear envelope disintegrates, and chromatin condenses. Metaphase is characterized by chromosome alignment on the metaphase plate, while sister chromatids migrate toward opposite poles of the cell during anaphase. During telophase, the nuclear envelope reforms and chromatin reverts to its uncondensed form.

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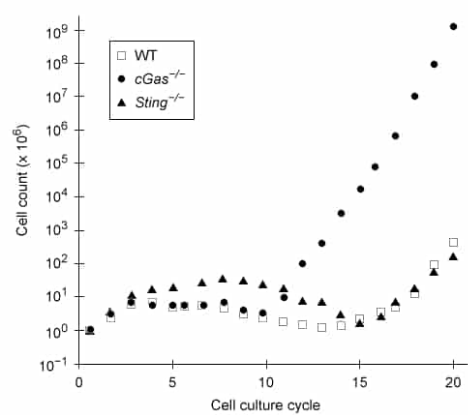


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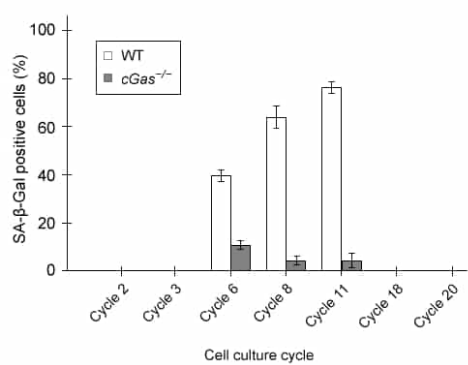


Figure 2 Percentage of *cGas*^{-/-} and WT cells positive for SA-β-Gal

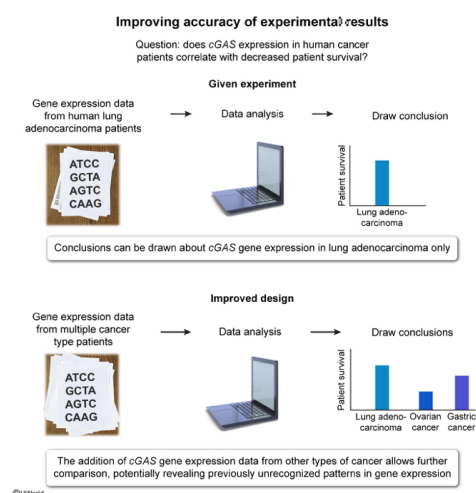
In an additional experiment, researchers seek to determine whether *cGAS* expression in human cancer patients is correlated with decreased patient survival. As one step in this experiment, the researchers analyzed gene expression data from a publicly available dataset relating to adult lung adenocarcinoma patients. Which change to this protocol would increase the accuracy of results?

- ☐ A. Limit the portion of the dataset analyzed to individuals of one gender. [9%]
- ☐ B. Include survival data pertaining to children with lung adenocarcinoma. [13%]
- ☒ C. Include gene expression datasets relating to other types of cancers. [58%]
- ☐ D. Limit the portion of the dataset analyzed to individuals living in the same location. [19%]

Correct

57% answered correctly

Explanation:



Tumor suppressor genes regulate DNA repair by pausing or repressing the **cell cycle**. This regulation helps ensure that only normal cells finish cell division. Programmed cell death (**apoptosis**) is induced in cells that cannot be repaired. In cancer cells, tumor suppressor genes are inhibited, allowing uncontrolled progression of the cell cycle and **tumor** formation and growth.

In this question, researchers seek to determine whether cGAS expression in human cancer patients is correlated with decreased patient survival. These researchers analyzed gene expression data from a dataset relating to adult lung adenocarcinoma patients to assist in elucidating the role of cGAS expression in cancer patient survival.

To improve the accuracy of the experimental results, the researchers could include **gene expression datasets** relating to **other types of cancers**. This would allow analysis of the role of cGAS in different types of cancer cells, rather than only lung adenocarcinoma cancer cells. Having this comparison data may reveal previously unrecognized patterns in cGAS expression in cancer cells, or it may provide evidence that there is no correlation between cGAS expression and survival across multiple cancer types.

The other choices would *not* increase the accuracy of the experimental results.

(Choices A and D) Limiting the dataset analyzed to individuals of one gender or those living in the same location would *limit* the scope of the experimental results.

(Choice B) The researchers are attempting to elucidate the role of cGAS in cancer patient survival. Including survival data pertaining to children with lung adenocarcinoma in addition to adult data would serve only to highlight any potential differences in cGAS expression between adults and children, not the overall role of cGAS in cancer patient survival.

Educational objective:

Tumor suppressor genes, which are able to induce programmed cell death, are often inhibited in cells that are cancerous.

Time spent: 0 sec

Subject:
Biology

QID: 403808

System:
Reproduction

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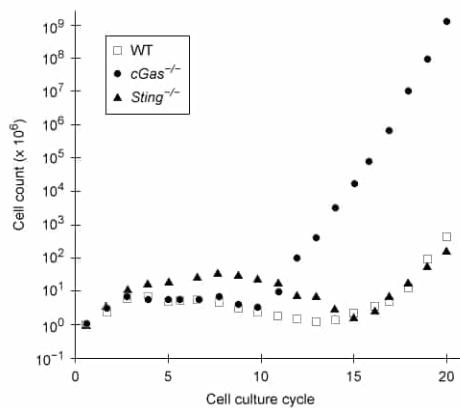


Figure 1 Effect of deletion of cGAS or STING on immortalization in MEFs

Experiment 2

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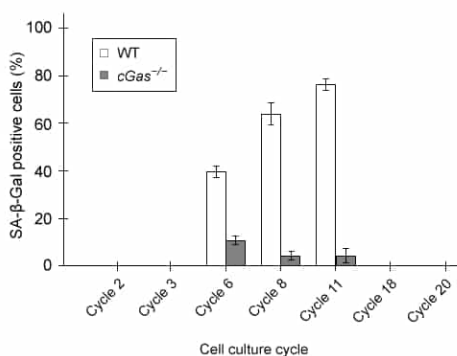


Figure 2 Percentage of $cGas^{-/-}$ and WT cells positive for SA- β -Gal

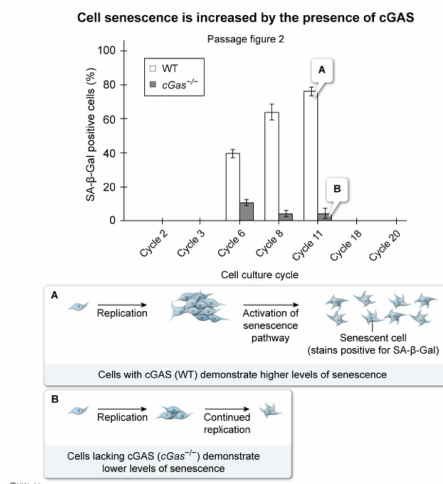
From the data in Figure 2, researchers can conclude that cell senescence is:

- ☐ A. decreased by the presence of cGAS. [29%]
- ☒ B. increased by the presence of cGAS. [66%]
- ☐ C. increased at least 2 times by the absence of STING. [1%]
- ☐ D. significantly higher at culture cycle 6 than at cycle 11 in WT cells. [2%]

Correct

65% answered correctly

Explanation:



Senescence is a natural aging process during which cells and tissues acquire damage (eg, [mutations](#)), causing diminished function over time. In cells, senescence occurs when [cell cycle](#) progression no longer occurs.

In Experiment 2, researchers explored the role of cGAS in senescence phenotypes in MEFs during their spontaneous immortalization (extended proliferation). Cells lacking cGAS (cGas^{-/-}) and wild-type (WT) cells with SA-β-Gal were stained, a commonly used method to detect whether cells are senescent. Figure 2 shows that when SA-β-Gal positive cells are detected, there is a significantly higher percentage of these cells in WT than in cGas^{-/-} cells.

These results suggest that when cGAS is not present, there are *fewer* senescent cells. Likewise, when cGAS is present (ie, in WT cells), there are *more* senescent cells. Therefore, researchers can conclude that **cell senescence is increased by the presence of cGAS**.

(Choice A) Figure 2 shows that cell senescence is increased (*not* decreased) by the presence of cGAS.

(Choice C) Figure 1 (*not* Figure 2) provides data on STING. Figure 2 demonstrates senescence data only in cells lacking cGAS, *not* STING.

(Choice D) Cell senescence is [significantly lower](#) (*not* higher) at cell culture cycle 6 than at cycle 11 in WT cells, according to Figure 2.

Educational objective:

Senescence is an aging process that occurs when cell cycle progression no longer occurs.

Time spent: 0 sec

Subject:
Biology

QID: 403809

System:
Reproduction